

UR5: Forward Kinematics

RBE 501

1 Product of Exponentials (POE) Forward Kinematics

In the space-frame Product of Exponentials (POE) approach, the continuous configuration state of the end-effector relative to the base frame is described by:

$$T(\theta) = e^{[S_1]\theta_1} e^{[S_2]\theta_2} e^{[S_3]\theta_3} e^{[S_4]\theta_4} e^{[S_5]\theta_5} e^{[S_6]\theta_6} M \quad (1)$$

Home Position Matrix (M). When all joint angles are zero ($\theta = \mathbf{0}$), the manipulator is in a fully extended vertical position. The initial end-effector pose matrix $M \in SE(3)$ is given by:

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & d_4 + d_6 \\ 0 & 0 & 1 & d_1 + a_2 + a_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

Joint 1. The rotation axis points along the positive Z -axis, passing through the origin.

$$\omega_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{q}_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \boxed{\mathcal{S}_1 = [0 \ 0 \ 1 \ 0 \ 0 \ 0]^T} \quad (3)$$

Joint 2. The axis is parallel to the Y -axis, offset by d_4 laterally and d_1 vertically.

$$\omega_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_2 = \begin{bmatrix} 0 \\ d_4 \\ d_1 \end{bmatrix} \Rightarrow \boxed{\mathcal{S}_2 = [0 \ 1 \ 0 \ d_1 \ 0 \ 0]^T} \quad (4)$$

Joint 3. The axis is parallel to the Y -axis, located at the upper arm height.

$$\omega_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_3 = \begin{bmatrix} 0 \\ d_4 \\ d_1 + a_2 \end{bmatrix} \Rightarrow \boxed{\mathcal{S}_3 = [0 \ 1 \ 0 \ d_1 + a_2 \ 0 \ 0]^T} \quad (5)$$

Joint 4. The axis is parallel to the Y -axis, located at the forearm length height.

$$\omega_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_4 = \begin{bmatrix} 0 \\ 0 \\ d_1 + a_2 + a_3 \end{bmatrix} \Rightarrow \boxed{\mathcal{S}_4 = [0 \ 1 \ 0 \ d_1 + a_2 + a_3 \ 0 \ 0]^T} \quad (6)$$

Joint 5. The axis is parallel to the Z -axis at home position, sharing the same vertical height as Joint 4.

$$\omega_5 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{q}_5 = \begin{bmatrix} 0 \\ d_4 \\ d_1 + a_2 + a_3 \end{bmatrix} \Rightarrow \boxed{\mathcal{S}_5 = [0 \ 0 \ 1 \ d_4 \ 0 \ 0]^T} \quad (7)$$

Joint 6. The terminal rotation axis is parallel to the Y -axis.

$$\omega_6 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{q}_6 = \begin{bmatrix} 0 \\ d_4 \\ d_1 + a_2 + a_3 + d_5 \end{bmatrix} \Rightarrow \boxed{\mathcal{S}_6 = [0 \ 1 \ 0 \ d_1 + a_2 + a_3 + d_5 \ 0 \ 0]^T} \quad (8)$$

2 Geometric Jacobian Formulation

The 6×6 Geometric Jacobian matrix $J(\mathbf{q})$ maps the joint velocities directly to the end-effector spatial velocities. It is constructed column by column based on the current axis orientation vectors $\mathbf{z}_{i-1}$ and link frame origins $\mathbf{o}_{i-1}$ extracted from the forward kinematics steps:

$$T_{01} = e^{[S_1]\theta_1} M, \quad T_{02} = T_{01} e^{[S_2]\theta_2}, \quad \dots, \quad T_{06} = T(\theta) \quad (9)$$

Let $\mathbf{o}_{end} = T_{06}(1 : 3, 4)$ represent the instantaneous position of the end-effector tip.

Jacobian Columns Mapping.

$$\text{Column 1 (Joint 1): } J_1 = \begin{bmatrix} \mathbf{z}_0 \times (\mathbf{o}_{end} - \mathbf{o}_0) \\ \mathbf{z}_0 \end{bmatrix}, \quad \text{where } \mathbf{z}_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \mathbf{o}_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (10)$$

$$\text{Column 2 (Joint 2): } J_2 = \begin{bmatrix} \mathbf{z}_1 \times (\mathbf{o}_{end} - \mathbf{o}_1) \\ \mathbf{z}_1 \end{bmatrix}, \quad \text{where } \mathbf{z}_1 = T_{01}(1 : 3, 3), \mathbf{o}_1 = T_{01}(1 : 3, 4) \quad (11)$$

$$\text{Column 3 (Joint 3): } J_3 = \begin{bmatrix} \mathbf{z}_2 \times (\mathbf{o}_{end} - \mathbf{o}_2) \\ \mathbf{z}_2 \end{bmatrix}, \quad \text{where } \mathbf{z}_2 = T_{02}(1 : 3, 3), \mathbf{o}_2 = T_{02}(1 : 3, 4) \quad (12)$$

$$\text{Column 4 (Joint 4): } J_4 = \begin{bmatrix} \mathbf{z}_3 \times (\mathbf{o}_{end} - \mathbf{o}_3) \\ \mathbf{z}_3 \end{bmatrix}, \quad \text{where } \mathbf{z}_3 = T_{03}(1 : 3, 3), \mathbf{o}_3 = T_{03}(1 : 3, 4) \quad (13)$$

$$\text{Column 5 (Joint 5): } J_5 = \begin{bmatrix} \mathbf{z}_4 \times (\mathbf{o}_{end} - \mathbf{o}_4) \\ \mathbf{z}_4 \end{bmatrix}, \quad \text{where } \mathbf{z}_4 = T_{04}(1 : 3, 3), \mathbf{o}_4 = T_{04}(1 : 3, 4) \quad (14)$$

$$\text{Column 6 (Joint 6): } J_6 = \begin{bmatrix} \mathbf{z}_5 \times (\mathbf{o}_{end} - \mathbf{o}_5) \\ \mathbf{z}_5 \end{bmatrix}, \quad \text{where } \mathbf{z}_5 = T_{05}(1 : 3, 3), \mathbf{o}_5 = T_{05}(1 : 3, 4) \quad (15)$$

The combined configuration formula for the matrix is expressed as:

$$\boxed{J(\mathbf{q}) = \begin{bmatrix} \mathbf{z}_0 \times (\mathbf{o}_{end} - \mathbf{o}_0) & \dots & \mathbf{z}_5 \times (\mathbf{o}_{end} - \mathbf{o}_5) \\ \mathbf{z}_0 & \dots & \mathbf{z}_5 \end{bmatrix}} \quad (16)$$